Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM :
To analyze the given dataset and perform data preprocessing and visualization.

Description: Use sample data set sales-data.csv

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
#loading the data into the dataframe
df = pd.read_csv('/SALES DATA - SALES DATA.csv')
df.head()
df.isnull().sum()
df.describe()

product_summary = df.groupby('Product').agg({
'Sales': 'sum',
'Quantity': 'sum'
}).reset_index()
product_summary

#bar chart

total_sales = df.groupby('Product')['Sales'].sum()

plt.figure(figsize=(8,5))
plt.bar(total_sales.index, total_sales.values)

plt.title('Total Sales by Product')
plt.xlabel('Product')
plt.ylabel('Total Sales')

plt.show()
```

```
df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')

# Create line plot
plt.figure(figsize=(10,5))

plt.plot(df['Date'], df['Sales'])

plt.title('Sales Over Time')
plt.xlabel('Date')
plt.ylabel('Sales')

plt.xticks(rotation=45)
plt.tight_layout()

pivot_table = pd.pivot_table(
    df,
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

pivot_table

import matplotlib.pyplot as plt
import seaborn as sns

correlation_matrix = df.corr(numeric_only=True)

print(correlation_matrix)

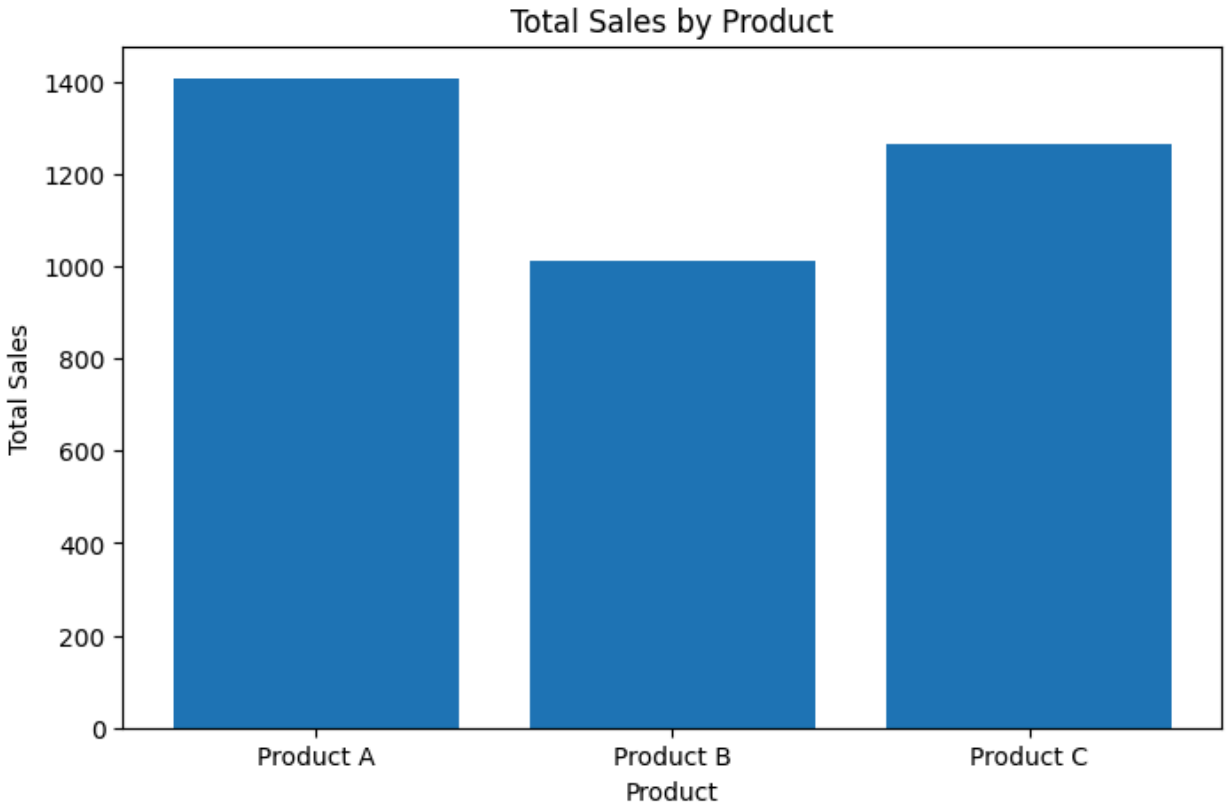
plt.figure(figsize=(6, 4))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f')

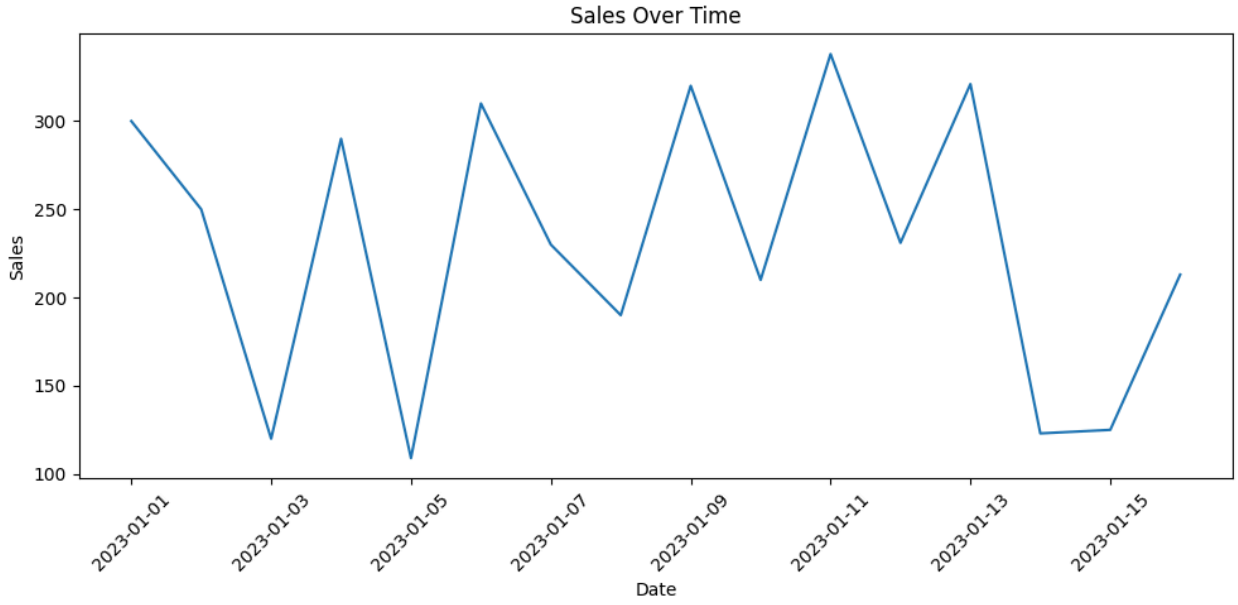
plt.title('Correlation Matrix')
plt.show()
```

SAMPLE OUTPUT:

	Sales	Quantity
count	16.00000	16.000000
mean	230.00000	5.187500
std	79.51939	1.905037
min	109.00000	2.000000
25%	173.75000	3.750000
50%	230.50000	5.000000
75%	302.50000	7.000000
max	338.00000	8.000000

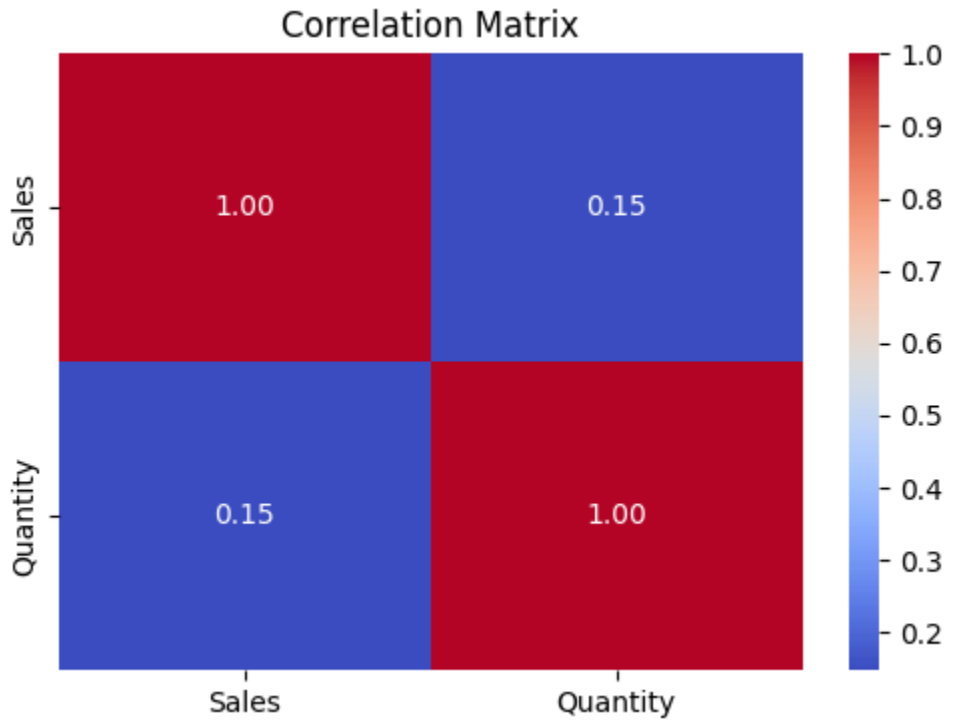
	Product	Sales	Quantity
0	Product A	1406	28
1	Product B	1010	26
2	Product C	1264	29





Product	Product A	Product B	Product C

Region			
East	0	0	1264
North	1406	0	0
South	0	563	0
West	0	447	0



RESULT:

The given dataset is analysed and visualized successfully.